

Proof. First we note that for a.e. t there exists, thanks to Theorem 5.14, a vector field $\mathbf{v}_t \in L^p(\mu_t)$ such that $\partial_t \mu_t + \nabla \cdot (\mathbf{v}_t \mu_t) = 0$, and with $\|\mathbf{v}_t\|_{L^p(\mu_t)} \leq |\mu'| (t)$. The same approximation argument (with the same convolution kernels) as in Theorem 5.14 provides a regularized curve of measures μ_t^ε and a regularized vector field $\mathbf{v}_t^\varepsilon$, with $\|\mathbf{v}_t^\varepsilon\|_{L^p(\mu_t^\varepsilon)} \leq \|\mathbf{v}_t\|_{L^p(\mu_t)}$. This vector field has a flow $Y : \Omega \rightarrow \text{AC}(\Omega)$, where $Y(x)$ is the curve $y_x(\cdot)$, and we know $\mu_t^\varepsilon = (Y_t)_\# \mu_0^\varepsilon$, where $Y_t(x) = Y(x)(t) = y_x(t)$. Let us also define the measure $Q^\varepsilon := Y_\# \mu_0^\varepsilon$. We have $\mu_t^\varepsilon = (e_t)_\# Q^\varepsilon$.

Let us compute $\int_{\mathcal{C}} K_p dQ^\varepsilon$:

$$\begin{aligned} \int_{\mathcal{C}} K_p(\omega) dQ^\varepsilon(\omega) &= \int_{\Omega} \int_0^1 |\mathbf{v}_t^\varepsilon(y_x(t))|^p dt d\mu_0^\varepsilon(x) = \int_0^1 \int_{\Omega} |\mathbf{v}_t^\varepsilon|^p d\mu_t^\varepsilon \\ &\leq \int_0^1 \int_{\Omega} |\mathbf{v}_t|^p d\mu_t = \int_0^1 |\mu'| (t)^p dt. \end{aligned}$$

Now (as we saw many times in Chapter 4), this implies that the measures Q^ε are tight, since there is a bound on $\int_{\mathcal{C}} K_p dQ^\varepsilon$ and K_p is such that

$$\{\omega \in \text{AC}(\Omega) : K_p(\omega) \leq L\}$$

is compact in $C^0([0, 1])$ (for the uniform convergence) for every L . Also, K_p is l.s.c. for the same convergence. Hence, we can obtain the existence of a subsequence such that $Q^\varepsilon \rightharpoonup Q$ and $\int_{\mathcal{C}} K_p dQ \leq \liminf_{\varepsilon} \int_{\mathcal{C}} K_p dQ^\varepsilon$. From $\mu_t^\varepsilon = (e_t)_\# Q^\varepsilon$ we get $\mu_t = (e_t)_\# Q$ and we conclude. $\square$

Proposition 5.32. *Let $(\mu_t)_t$ be a constant speed geodesic in $\mathbb{W}_p(\Omega)$ between μ and ν and suppose $p > 1$. Then there exists an optimal $\gamma \in \Pi(\mu, \nu)$ for the transport cost $c(x, y) = |x - y|^p$ such that for every $t \in [0, 1]$ we have $\mu_t = (\pi_t)_\# \gamma$.*

Proof. Since the curve is a constant speed geodesic, we have $|\mu'| (t) = W_p(\mu, \nu)$ for a.e. t (the metric derivative being computed according to W_p). Applying our previous Proposition 5.31, we get the existence of a measure $Q \in \mathcal{P}(\mathcal{C})$ such that $\mu_t = (e_t)_\# Q$ and $\int_{\mathcal{C}} K_p(\omega) dQ(\omega) \leq \int_0^1 |\mu'| (t)^p dt$.

From Jensen inequality we have $|\omega(0) - \omega(1)|^p \leq K_p(\omega)$, with equality if and only if ω is a segment parametrized with constant speed. Hence, we have

$$\begin{aligned} W_p^p(\mu, \nu) &\leq \int_{\mathcal{C}} |\omega(0) - \omega(1)|^p dQ(\omega) \\ &\leq \int_{\mathcal{C}} K_p(\omega) dQ(\omega) \leq \int_0^1 |\mu'| (t)^p dt \leq W_p^p(\mu, \nu). \end{aligned}$$

The first inequality comes from $\gamma := (e_0, e_1)_\# Q \in \Pi(\mu_0, \mu_1)$; the second is a consequence of the above inequality for K_p ; the last comes from the properties of Q . Thus, all inequalities are equalities. Hence Q is concentrated on curves which are constant speed segments and γ is an optimal transport plan. This proves $\mu_t = (\pi_t)_\# \gamma$. $\square$

Note how the above proof is reminiscent of that of Theorem 4.13, with here a dynamical framework, compared to the static framework in Chapter 4.

It is not difficult to see that both Proposition 5.31 and 5.32 do not hold for $p = 1$. The counter-example is, as usual, given by $\mu_t = (1 - t)\delta_{x_0} + t\delta_{x_1}$.

We finish this section insisting on the role that Wasserstein geodesics play in applications, as they allow to build an interesting interpolation between two distribution of masses (or two histograms), moving slowly the particles from one distribution to the other. For instance, Figure 5.2, shows the interpolation between the chrominance histograms of Figure 2.4. These intermediate histograms could also be used to produce intermediate images.



Figure 5.2: Nine time steps of the interpolation (top, left to right, then bottom, left to right), obtained via a $\mathbb{W}_2$ geodesic curve, between the histograms on the 2D space of chroma signals of the two images of Figure 2.4. Pictures kindly provided by G. Peyré.

5.5 Discussion

5.5.1 The W_∞ distance

The case $p = \infty$ in the definition of W_p deserves special attention. It is easy to guess how to define a W_∞ distance on probability measures:

$$W_\infty(\mu, \nu) := \inf \{ \|x - y\|_{L^\infty(\gamma)} : \gamma \in \Pi(\mu, \nu) \},$$

where the minimization problem in the right-hand side is exactly the one studied in Section 3.2. As we did in the rest of this chapter, the difference here is that, instead of paying attention to the optimal map, we mainly consider the optimal cost as a function of the two measures, and use it as a distance on the space of probabilities.

In order to properly define $W_\infty(\mu, \nu)$, one should restrict to $\mathcal{P}_\infty(\Omega)$, which is the space of measures with bounded support. Otherwise, there could be pairs of measures such that no bounded-displacement plan γ exists. From the results in Section 3.2 we know that, if $\mu \ll \mathcal{L}^d$, then $W_\infty(\mu, \nu)$ is also equal to

$$\inf \{ \|T(x) - x\|_{L^\infty(\mu)} : T_\# \mu = \nu \}.$$

It is not difficult to check the expected property

$$W_\infty(\mu, \nu) = \lim_{p \rightarrow \infty} W_p(\mu, \nu),$$

where the right-hand-side is increasing in p (the proof is proposed as an exercise, Ex(35)). The triangle inequality can be proved by passing to the limit in p in the corresponding triangle inequality for W_p .

Note that, in dimension one, as a consequence of the results of Chapter 2, the optimal way to displace particles is the monotone increasing map, for every cost $|x - y|^p$ with $p < \infty$. Passing to the limit, this is also the optimal map for the L^∞ problem.

In general, we have a new distance on probability measures on compact sets, and it measures the minimal maximal displacement that should be done to move particles from one distribution to the other. It is interesting to see that, though the modeling which gives rise to this distance is the same as in the case $p < \infty$, the topology induced by it is not at all the same.

Indeed, if one takes two different points $x, y \in \Omega$ and two different values $s, t \in [0, 1]$, one can easily check $W_\infty((1-t)\delta_x + t\delta_y, (1-s)\delta_x + s\delta_y) = |x - y|$. In particular, the probabilities $(1-t)\delta_x + t\delta_y$ do not tend to δ_x as $t \rightarrow 0$, even if there is convergence for all the distances W_p , and even for the strong convergence of measures. As a matter of fact, even if Ω is compact, the space $(\mathcal{P}(\Omega), W_\infty)$ is neither compact, nor separable, as soon as the cardinality of Ω is larger than 1.

On the other hand, the characterization of Lipschitz curves via the existence of a velocity field $\mathbf{v}_t$, satisfying $\|\mathbf{v}_t\|_{L^\infty(\mu_t)} \leq |\mu'|_{W_\infty}(t)$ stays true, and can be easily deduced from Theorem 5.14 as a limit $p \rightarrow \infty$. This fact, together with an extended use of Lipschitz curves for W_∞ , has been used in [80] in order to analyze an alternative model for branched transport. If the role of this model is very limited, some examples and considerations in the paper are useful to understand the structure of Wasserstein spaces.

However, the distance W_∞ appears naturally in many modeling issues, and happens to be very useful to analyze very different phenomena. One of the first papers using such a distance was [232], devoted to the shape of rotating stars. In such a paper, the choice of W_∞ was motivated by the following observation (that we will develop better in Section 7.4.3): a measure μ is a local minimizer in $\mathbb{W}_\infty$ of some functional F if and only if every point $x \in \text{spt}(\mu)$ satisfies a local minimality condition for a function f (equal to the first variation of F , see Section 7.2). This equivalence fails if we replace W_∞ with W_p .

Finally, it is interesting to mention a reverse inequality between W_∞ and W_p (which could not be deduced from Equation (5.1)), of the form

$$W_\infty(\mu, \nu) \leq C W_p^{\frac{p}{p+d}}(\mu, \nu),$$

where the constant C depends on p , on the dimension d , on the domain Ω , and on a lower bound on the density of μ . This inequality is a consequence of a stronger result proved in [75]: if $\mu = f \cdot \mathcal{L}^d$, $f \geq a > 0$ and T is the optimal transport map from μ to another measure ν , then we have

$$\|T - \text{id}\|_{L^\infty}^{p+d} \leq C(\Omega, p, d) \int_\Omega |T(x) - x|^p dx \leq \frac{C(\Omega, p, d)}{a} \int_\Omega |T(x) - x|^p d\mu(x).$$

This is proven by using the c -cyclical monotonicity condition on T so as to guarantee that, whenever we fix a point x with $|T(x) - x| > 0$, there exists a certain region R around x , with volume proportional to $|T(x) - x|^d$, such that for every $y \in R$ the distance $|T(y) - y|$ is not smaller than a certain fraction of $|T(x) - x|$.

5.5.2 Wasserstein and negative Sobolev distances

We already noted that the distance W_1 is induced by a norm, which is the dual of the Lipschitz seminorm. Indeed, if we consider the space $\mathcal{X}$ of Lipschitz function with zero mean, endowed with the norm $\|f\| := \text{Lip}(f)$, the distance W_1 on $\mathcal{P}_1(\Omega)$ is exactly the distance associated to the norm of the topological dual $\mathcal{X}'$. It is interesting to note that the space of probability measures with this distance is complete, but the space of finite measures endowed with the same norm is not. More precisely, one can consider zero-mean finite scalar measures endowed with the norm $\|\rho\| := \sup\{\int f d\rho : f \in \text{Lip}_1(\Omega)\}$ and look at its completion. Note that the sequence $\rho_n := \sum_{i=1}^n \delta_{1/2i} - \delta_{1/(2i-1)}$ is Cauchy for this norm, but its limit is an infinite sum of Dirac masses, which is not a finite scalar measure. The completion of the set of measures for this norm has been studied in [73], where the authors prove that it amounts to the set of distributions which are the divergence of an L^1 vector field. This nice result makes an interesting link with the L^1 -Beckmann problem studied in Chapter 4.

Anyway, once the role of the distance W_1 in the duality with $W^{1,\infty}$ functions is clear, one could wonder whether other distances W_p are in duality with Sobolev spaces of the form $W^{1,q}$ (for instance, with $q = p' = p/(p-1)$). The answer is obviously “not”, since not all probability measures belong to the dual space $W^{-1,p}$ (in particular, Dirac masses only belong to the dual of $W^{1,q}$ when functions in $W^{1,q}$ are continuous, i.e. when $q > d$). Yet, some partial answers and some analogies exist. For instance the following lemma was both proven in [215] and later in [224] for different purposes.

Lemma 5.33. *Assume that μ and ν are absolutely continuous measures on a convex domain Ω , whose densities are bounded by the same constant C . Then, for all function $\phi \in H^1(\Omega)$, we have the following inequality:*

$$\int_{\Omega} \phi d(\mu - \nu) \leq \sqrt{C} \|\nabla \phi\|_{L^2(\Omega)} W_2(\mu, \nu)$$

Proof. Let μ_t be the constant speed geodesic between μ and ν , and let $\mathbf{v}_t$ be the velocity field associated to this curve by Theorem 5.14. Then (μ, ν) satisfies the continuity equation, and $\|\mathbf{v}_t\|_{L^2(\mu_t)} = W_2(\mu, \nu)$. We use the fact that μ_t is absolutely continuous for all t , and its density is bounded by the same constant C , which will be proven in Chapter 7 (Prop 7.29). Therefore:

$$\begin{aligned} \int_{\Omega} \phi d(\mu - \nu) &= \int_0^1 \frac{d}{dt} \left(\int_{\Omega} \phi(x) d\mu_t(x) \right) dt = \int_0^1 \int_{\Omega} \nabla \phi \cdot \mathbf{v}_t d\mu_t dt \\ &\leq \left(\int_0^1 \int_{\Omega} |\nabla \phi|^2 d\mu_t dt \right)^{1/2} \left(\int_0^1 \int_{\Omega} |\mathbf{v}_t|^2 d\mu_t dt \right)^{1/2} \\ &\leq \sqrt{C} \|\nabla \phi\|_{L^2(\Omega)} W_2(\mu, \nu), \end{aligned}$$

and the proof is completed. $\square$ $\square$

We do not develop here the case of different exponents than $p = p' = 2$, since the quadratic case seems to be the most interesting (see **Ex(38)** for the general case). The exact negative Sobolev norm appearing in the quadratic case is the $\dot{H}^{-1}$ norm, which deserves some remarks. First let us define

$$\|\mu - \nu\|_{\dot{H}^{-1}(\Omega)} = \sup \left\{ \int \phi \, d(\mu - \nu) : \|\nabla \phi\|_{L^2} \leq 1 \right\}.$$

Note that we also have $\|\mu - \nu\|_{\dot{H}^{-1}(\Omega)} = \|\nabla u\|_{L^2}$, where u is the solution of the Neumann problem

$$\begin{cases} -\Delta u = \mu - \nu, & \text{in } \Omega, \\ \frac{\partial u}{\partial \mathbf{n}} = 0 & \text{on } \partial\Omega. \end{cases}$$

Indeed, for this choice of u we have $\int \phi \, d(\mu - \nu) = \int \nabla \phi \cdot \nabla u \leq \|\nabla \phi\|_{L^2} \|\nabla u\|_{L^2}$, with equality for $\phi = u / \|\nabla u\|_{L^2}$. We can also see that we have

$$\|\mu - \nu\|_{\dot{H}^{-1}(\Omega)} = \min \{ \|\mathbf{w}\|_{L^2} : \nabla \cdot \mathbf{w} = \mu - \nu \},$$

since the optimal $\mathbf{w}$ in this minimization problem is a gradient (see Section 4.4.1 and the considerations on Problem (4.27)), and we have $\mathbf{w} = -\nabla \psi$.

The interest for the relation between the distances W_2 and $\dot{H}^{-1}$ (where $\dot{H}^{-1}$ denotes the dual of the space $\mathcal{X}$ of zero-mean H^1 functions on a connected domain, endowed with the L^2 norm of its gradient) is now very lively because these two distances are asymptotically equivalent when the involved densities are close to the value $\rho = 1$, which is the relevant case when dealing with evolution problems where congestion imposes an L^∞ constraint $\rho \leq 1$, as in Crowd Motion (see Section 8.4.2). These asymptotical equivalence can be made precise through some general estimates. We cite a short note by R. Peyre ([251]) where some universal estimates are proven, and a funny equality in dimension 1 (see **Ex(64)**). Moreover, we also prove the following result.

Theorem 5.34. *Assume that μ and ν are absolutely continuous measures on a convex domain Ω , with densities bounded from below and from above by two (same) constants a, b with $0 < a < b < +\infty$. Then, we have*

$$b^{-1/2} \|\mu - \nu\|_{\dot{H}^{-1}(\Omega)} \leq W_2(\mu, \nu) \leq a^{-1/2} \|\mu - \nu\|_{\dot{H}^{-1}(\Omega)}.$$

Proof. The inequality $b^{-1/2} \|\mu - \nu\|_{\dot{H}^{-1}(\Omega)} \leq W_2(\mu, \nu)$ is a consequence of Lemma 5.33 and of the definition of the $\dot{H}^{-1}$ norm.

To prove the opposite inequality can be obtained through the Benamou-Brenier formula of Theorem 5.28: consider any vector field $E \in \mathcal{M}_{\text{div}}^d(\Omega)$ with $\nabla \cdot E = \mu - \nu$ and $\rho_t := (1-t)\mu + t\nu$. The pair (ρ_t, E_t) , where $E_t = E$ is taken independent of time, is admissible in the Benamou-Brenier formula. If $E \in L^2(\Omega; \mathbb{R}^d)$, we have

$$W_2^2(\mu, \nu) \leq \mathcal{B}_2(\rho, E) = \int_0^1 \int_\Omega \frac{|E|^2}{\rho_t} \leq \frac{1}{a} \int_\Omega |E|^2,$$

where the last inequality is justified by $\rho_t = (1-t)\mu + t\nu \geq (1-t)a + ta = a$. Since E is arbitrary with the only constraint $\nabla \cdot E = \mu - \nu$, we obtain

$$W_2^2(\mu, \nu) \leq \frac{1}{a} \min \{ \|\mathbf{w}\|_{L^2}^2 : \nabla \cdot \mathbf{w} = \mu - \nu \} = \frac{1}{a} \|\mu - \nu\|_{H^{-1}(\Omega)}^2,$$

which gives the second part of the statement. $\square$ $\square$

5.5.3 Wasserstein and branched transport distances

Another class of distances which has been compared to the Wasserstein distances is that induced by branched transport. We recall that $d_\alpha(\mu, \nu)$ is defined as the minimal value of the branched transport cost (see Section 4.4.2) between two measures μ, ν , as soon as $\alpha > 1 - \frac{1}{d}$. In this case it is a distance, and it metrizes the weak convergence of probabilities on a bounded domain.

In branched transport the cost per unit length to move a mass m is given by m^α , and we can assume $m \leq 1$ (because there are no cycles in the optimal networks). Hence, one immediately remarks $m^\alpha \geq m$, which implies $d_\alpha \geq d_1$, and the distance d_1 is nothing but W_1 .

Since the two distances d_α and W_1 satisfy an inequality, and they induce the same topology, it was a question raised by C. Villani whether the two distances are metrically equivalent, i.e. we can bound above d_α with W_1 . Inequalities of the form $d_\alpha \leq CW_1$ are necessarily false, since they do not scale in the correct way w.r.t. masses. Indeed, even if we stick to probability measures, we can consider the following situation: take $\mu, \nu \in \mathcal{P}(\Omega_1)$ and a third measure ρ concentrated on another domain Ω_2 , disjoint from Ω_1 . Then take $\mu_\varepsilon = (1-\varepsilon)\rho + \varepsilon\mu$ and $\nu_\varepsilon = (1-\varepsilon)\rho + \varepsilon\nu$. Since both W_1 and d_α only depend on the difference between the two measures (which is false for W_p , with $p > 1$), by a scaling argument it is clear that we have $W_1(\mu_\varepsilon, \nu_\varepsilon) = c_1\varepsilon$ and $d_\alpha(\mu_\varepsilon, \nu_\varepsilon) = c_\alpha\varepsilon^\alpha$. This rules out the possibility of a linear bound, but leaves the possibility of an inequality of the form $d_\alpha \leq CW_1^\beta$ for $\beta \leq \alpha$.

This question has been analyzed in [240], where the sharp exponent has been found, $\beta = d(\alpha - (1 - \frac{1}{d}))$. Indeed, a simple example shows that it cannot be improved.

Example 5.35 (Optimal exponent for $d_\alpha \leq W_1^\beta$). Divide the cube $[0, 1]^d$ into n^d small cubes of edge $1/n$ and set $\mu_n = \sum_{i=1}^{n^d} \frac{1}{n^d} \delta_{x_i}$ and $\nu_n = \sum_{i=1}^{n^d} \frac{1}{n^d} \delta_{y_i}$, where each x_i is a vertex of one of the n^d cubes (let us say the vertex with minimal sum of the d coordinates) and the corresponding y_i is the center of the same cube. In this way the point y_j closest to x_i is exactly y_i . Thus the optimal configuration both for d_α and W_1 is obtained by linking any x_i directly to the corresponding y_i , and we have

$$\begin{aligned} d_\alpha(\mu_n, \nu_n) &= n^d \left(\frac{1}{n^d} \right)^\alpha \frac{c}{n} = cn^{-d(\alpha - (1 - 1/d))} \\ W_1(\mu_n, \nu_n) &= n^d \frac{1}{n^d} \frac{c}{n} = cn^{-1} \end{aligned}$$

The proof of the inequality $d_\alpha \leq CW_1^{d(\alpha - (1 - 1/d))}$ is a bit technical. Yet, in order to give an idea, we just sketch an alternative proof (from [81]), which provides a slightly

weaker inequality, namely $d_\alpha \leq CW_p^{d(\alpha-(1-1/d))}$ for $p = 1/\alpha$. This inequality is also optimal in some sense, as it is also possible to strengthen the lower bound $d_\alpha \geq W_1$ turning it $d_\alpha \geq W_{1/\alpha}$ (this has been first remarked in [221] and then in [81]). The choice of the exponent p is very natural because of scaling reasons. Indeed, W_p scales as $m^{\frac{1}{p}}$ w.r.t. the mass (in the example above, if Ω_1 and Ω_2 are far enough so that the measure ρ becomes useless, then we have $W_p(\mu_\varepsilon, \nu_\varepsilon) = c\varepsilon^{\frac{1}{p}}$), which makes the choice $p = 1/\alpha$ the good one.

Proposition 5.36. *Let μ, ν be two probabilities on $[0, 1]^d$ and $\alpha > 1 - \frac{1}{d}$, $p = 1/\alpha$. Then*

$$W_p \leq d_\alpha \leq CW_p^{d(\alpha-(1-1/d))}.$$

Proof. First we prove the inequality $W_p \leq d_\alpha$. This is a consequence of (4.33). Indeed, W_p is a distance, it is l.s.c. for $\rightarrow$, and we have

$$\sum_{ij} |x_i - y_j| \gamma(\{(x_i, y_j)\})^\alpha \geq \left(\sum_{ij} |x_i - y_j|^p \gamma(\{(x_i, y_j)\}) \right)^\alpha.$$

This inequality is due to the choice $p = 1/\alpha$ and to the sub-additivity of $t \mapsto t^\alpha$. Passing to the minimum over $\gamma \in \Pi(\mu, \nu)$, we get $\bar{d}_\alpha(\mu, \nu) \geq W_p(\mu, \nu)$, whenever μ and ν are atomic. This implies $d_\alpha \geq W_p$.

In order to prove the other inequality, we divide the cube into n^d small disjoint cubes Q_i of edge $1/n$ and set

$$\mu_n = \sum_{i=1}^{n^d} \frac{1}{n^d} \mu(Q_i) \delta_{y_i} \quad \text{and} \quad \nu_n = \sum_{i=1}^{n^d} \nu(Q_i) \delta_{y_i},$$

where each point y_i is the center of the cube Q_i . Since the branched transport from $\mu \llcorner Q_i$ to $\mu(Q_i) \delta_{y_i}$ costs no more than $C\mu(Q_i)^{\alpha \frac{1}{n}}$ (see Proposition 4.38), we can estimate

$$\begin{aligned} d_\alpha(\mu, \nu) &\leq d_\alpha(\mu, \mu_n) + d_\alpha(\mu_n, \nu_n) + d_\alpha(\nu_n, \nu) \\ &\leq \frac{C}{n} \left(\sum_i \mu(Q_i)^\alpha + \sum_i \nu(Q_i)^\alpha \right) + d_\alpha(\mu_n, \nu_n) \leq \frac{C}{n} n^{d(1-\alpha)} + d_\alpha(\mu_n, \nu_n), \end{aligned}$$

where we used the inequality $\sum_{j=1}^k a_j^\alpha \leq k(\frac{1}{k})^\alpha$, which is valid whenever $\sum_{j=1}^k a_j = 1$ and $\alpha \in]0, 1[$ (it is just a version of Jensen inequality).

Then, let $\gamma \in \Pi(\mu_n, \nu_n)$ be an optimal transport plan for the cost $|x - y|^p$. We have $\gamma = \sum_{i,j} \gamma_{ij} \delta_{(x_i, x_j)}$ and we can decide to use segments connecting each point x_i to each point x_j with mass γ_{ij} as a (non-optimal) branched transport network. Its cost is $\sum_{i,j} \gamma_{ij}^\alpha |x_i - x_j|$. We denote by K the number of pairs (i, j) such that $\gamma_{ij} > 0$. Since γ is optimal in a linear programming problem, it may be taken extremal in the polyhedron of matrices $\{(\eta_{ij})_{i,j} : \eta_{ij} \geq 0, \sum_i \eta_{ij} = \nu(Q_j), \sum_j \eta_{ij} = \mu(Q_i)\}$. For these matrices, it is well-known that the number of non-zero entries does not exceed the sum of the

number of rows and columns (left as an exercise, **Ex(41)**), here $2n^d$. Hence we have

$$\begin{aligned} d_\alpha(\mu_n, \nu_n) &\leq \sum_{i,j} \gamma_{ij}^\alpha |x_i - x_j| \leq \left(\sum_{i,j} \gamma_{ij} |x_i - x_j|^p \right)^{\frac{1}{p}} (2n^d)^{1-\frac{1}{p}} \\ &= CW_p(\mu_n, \nu_n) n^{d(1-\alpha)} \leq C \left(W_p(\mu, \nu) + \frac{2}{n} \right) n^{d(1-\alpha)}, \end{aligned}$$

where we used the triangle inequality on W_p together with $W_p(\mu, \mu_n), W_p(\nu, \nu_n) \leq 1/n$. By putting together the estimates one gets

$$d_\alpha(\mu, \nu) \leq C n^{d(1-\alpha)-1} + C n^{d(1-\alpha)} W_p(\mu, \nu)$$

and we can conclude by taking $n \approx W_p(\mu, \nu)^{-1}$. $\square$ $\square$

5.5.4 The Sliced Wasserstein distance

Starting from the considerations of Section 2.5.2 it is possible, and convenient in many cases, to define a distance, which is alternative to the usual W_2 distance, based on the behavior of the measures “direction by direction”.

The idea is the following, and is detailed in [256]: given two measures $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, we define

$$SW_2(\mu, \nu) := \left(\int_{S^{d-1}} W_2^2((\pi_e)_\# \mu, (\pi_e)_\# \nu) d\mathcal{H}^{d-1}(e) \right)^{1/2},$$

where $\pi_e : \mathbb{R}^d \rightarrow \mathbb{R}$ is the projection on the axis directed according to the unit vector e , namely $\pi_e(x) = x \cdot e$, and $\mathcal{H}^{d-1}$ is the surface measure on S^{d-1} . This quantity could have been called “projected Wasserstein distance” (as it is based on the behavior through projections), but since in [256] it is rather called “sliced Wasserstein distance”, we prefer to keep the same terminology.

The fact that SW_2 is a distance comes from W_2 being a distance. The triangle inequality may be proven using the triangle inequality for W_2 (see Section 5.1) and for the L^2 norm. Positivity and symmetry are evident. The equality $SW_2(\mu, \nu) = 0$ implies $W_2^2((\pi_e)_\# \mu, (\pi_e)_\# \nu) = 0$ for all $e \in S^{d-1}$. This means $(\pi_e)_\# \mu = (\pi_e)_\# \nu$ for all e and it suffices (see Box 2.4 in Section 2.5.2 on Radon and X-ray transforms) to prove $\mu = \nu$.

It is evident from its definition, and from the fact that the maps π_e are 1-Lipschitz (hence $W_2^2((\pi_e)_\# \mu, (\pi_e)_\# \nu) \leq W_2^2(\mu, \nu)$) that we have $SW_2(\mu, \nu) \leq W_2(\mu, \nu)$. Moreover, the two distances also induce the same topology, at least on compact sets. Indeed, the identity map from $\mathbb{W}_2$ to $(\mathcal{P}(\Omega), SW_2)$ is continuous (from $SW_2 \leq W_2$) and bijective. Since the space where it is defined is compact, it is also a homeomorphism.

One can also prove more, i.e. an inequality of the form $W_2 \leq CSW_2^\beta$ for a suitable exponent $\beta \in]0, 1[$. Chapter 5 in [66] proves this inequality⁷ with $\beta = (2(d+1))^{-1}$.

⁷Indeed, one can define Sliced Wasserstein distances associated to other exponents p , and [66] proves $W_1 \leq CSW_1^{1/(d+1)}$; then, one can compose with $W_2 \leq CW_1^{1/2}$ to obtain the result for $p = 2$.

The interest in the use of this distance is the fact that one has a distance on $\mathcal{P}(\Omega)$ with very similar qualitative properties as W_2 , but much easier to compute, since it only depends on one-dimensional computations (obviously, the integral over $e \in \mathbb{S}^{d-1}$ is discretized in practice, and becomes an average over a large number of directions). For instance it is useful when the Wasserstein distance is used to evaluate distances between mass distributions (or images, where each image is associated to a measure on $\mathbb{R}^5$ by associating to each pixel its two spatial coordinates and its three color components), or when one look at the “horizontal” barycenter of several distributions, as in Section 5.5.5 here below (in the Euclidean space, the barycenter of $x_1, \dots, x_N$ is the point x which minimizes $\sum_i |x - x_i|^2$; in the Wasserstein space, the measure minimizing $\sum_i W_2^2(\rho, \rho_i)^2$ is the one which is located “at the middle” of $\rho_1, \dots, \rho_N$). In both cases, replacing W_2 with SW_2 simplifies a lot the computations. We remark anyway an important difference between W_2 and SW_2 : the latter is not a geodesic distance. On the contrary, the geodesic distance associated to SW_2 (i.e. the minimal length to connect two measures) is exactly W_2 .

We will also see in Section 8.4.2 that the gradient flow of SW_2^2 has an interest in itself and in the very next Section 5.5.5 we see the applications of SW_2^2 to barycenter computations.

5.5.5 Barycenters in $\mathbb{W}_2$

An interesting problem in applications, and in particular in image processing, is that of finding a reasonable notion of barycenter between two or more probability measures. Suppose that we have two B&W images, representing similar objects, but one being very dark (almost black) and one very pale (almost white). We denote by ρ^0 and ρ^1 their color histograms (see Section 2.5.1): what is the middle point between these two histograms? if we answer $(\rho^0 + \rho^1)/2$, then we obtain an image with some black and some white, while we could prefer to have an intermediate grey image. Also, we would like to provide a definition of middle point such that the middle point of δ_{x_0} and δ_{x_1} is $\delta_{x_{1/2}}$, with $x_{1/2} = (x_0 + x_1)/2$.

The notion of geodesic in Wasserstein spaces provides what we look for: just define the middle point as $\rho_{1/2}$, where ρ_t is the constant-speed geodesic between ρ^0 and ρ^1 , according to what we saw in Section 5.4. Also, we could look for a weighted barycenter, with weights $(1-t)$ and t , and the geodesic provides the correct answer. Some non-uniqueness issues subsist: geodesics are not always unique, and moreover they depend on the exponent p . We will soon fix the exponent issue by choosing $p = 2$, as we are going to see in a while.

What about a barycenter between three or more measures? if we want to make use of geodesics, we could define the barycenter of (ρ^0, ρ^1, ρ^2) , with weights $(1/3, 1/3, 1/3)$, as the barycenter with weights $(1/3, 2/3)$ between ρ^2 and the middle-point of ρ^0 and ρ^1 . Unfortunately, in the space $\mathbb{W}_2$ in dimension larger than 1, there is no associativity and this definition gives different results if we interchange the order of the measures ρ^i .

Hence, it is better to switch to another characterization of the barycenter in Euclidean spaces, which we will translate into the framework of Wasserstein spaces. We note that the barycenter with weights $\lambda_i \geq 0$ between points $x_i \in \mathbb{R}^d$ is the unique point

y minimizing $\sum_i \lambda_i |y - x_i|^2$. Hence, we could define the barycenter between ρ^i with weights λ_i as any measure solving

$$\min \left\{ \sum_{i=1}^N \lambda_i W_2^2(\rho, \rho^i)^2 : \rho \in \mathcal{P}(\Omega) \right\}. \quad (5.7)$$

Note that the same approach could be applied to general Riemannian manifolds, as the barycenter is otherwise not well-defined on non-linear structures. This explains also the choice of the exponent $p = 2$: using powers p of the distances we do not recover the usual barycenter in the Euclidean space (for $p = 1$ we have a notion of median), and using the square of a W_p distance is not very convenient. The uniqueness of the barycenter will be true as soon as one of the measures ρ^i is absolutely continuous (see, Proposition 7.19).

As an interesting example, it is not difficult to analyze the one-dimensional case. In this case, one can find the barycenter of some measures ρ^i with weights λ_i in the following way: pick one of these measures, say ρ^1 , define the optimal transport maps T_i sending ρ^1 onto ρ^i (i.e. the monotone maps, note that this would work for any exponent p), and set $\rho := (\sum_i \lambda_i T_i)_\# \rho^1$, where we set $T_1 = \text{id}$ (see also **Ex**(63)).

The minimization problem (5.7) has been first introduced in [1], where the authors also propose an interpretation in terms of a multi-marginal optimal transport problem. Indeed, one can choose as a variable, instead of $\rho \in \mathcal{P}(\Omega)$, a multi-marginal transport plan $\gamma \in \mathcal{P}(\Omega^{N+1})$, with $(\pi_i)_\# \gamma = \rho^i$ for $i = 1, \dots, N$ and $(\pi_0)_\# \gamma = \rho$ (the latter ρ being part of the unknown). The problem becomes

$$\min \int \left(\sum_{i=1}^N \lambda_i |x_i - x_0|^2 \right) d\gamma,$$

under the constraint that the marginals of γ for $i = 1, \dots, N$ are the ρ^i . The other marginal being arbitrary, one could decide to choose x_0 as a function of the points $(x_1, \dots, x_N)$, in an optimal way, i.e. $x_0 = \sum_i \lambda_i x_i$ (suppose $\sum_i \lambda_i = 1$). In this way we need to solve

$$\min \left\{ \int c(x_1, \dots, x_N) d\gamma : \gamma \in \Pi(\rho^1, \dots, \rho^N) \right\},$$

with $c(x_1, \dots, x_N) := \sum_{i=1}^N \lambda_i |x_i - \bar{x}|^2$ and $\bar{x} = \sum_{i=1}^N \lambda_i x_i$. Note that, in the standard case $\lambda_i = 1/N$, we have

$$c(x_1, \dots, x_N) = \frac{1}{N} \sum_{i=1}^N |x_i - \bar{x}|^2 = -|\bar{x}|^2 + \frac{1}{N} \sum_{i=1}^N |x_i|^2 = \frac{1}{2N^2} \sum_{i,j} |x_i - x_j|^2$$

(to be convinced of the last equality, just expand the squares). This shows that the problem is equivalent to the very first multi-marginal problem studied by Gangbo and Świąch in [178].

For the applications of barycenters in image processing we refer to [256], where, for the sake of numerical computations, the distance W_2 is replaced with SW_2 (see Section 5.5.4). An example is given in Figure 5.3. More recent numerical algorithms have also allowed to directly treat the W_2 distance, as is shown in Figure 5.4.

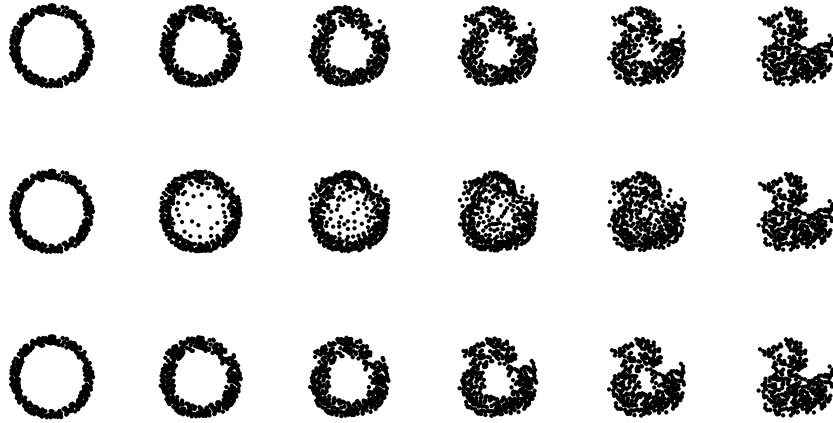


Figure 5.3: Interpolation between a point distribution on an annulus and on a duck. On the first line, an assignment is computed as in Figure 2.5.2, and then points move interpolating starting and arrival positions with constant speed. On the third line, the same is done with the optimal assignement, realizing the distance W_2 (hence we have a geodesic in $\mathbb{W}_2$). On the second line, the interpolations are computed as barycenters for SW_2 , minimizing the sum of the squared distances. Pictures kindly provided by J. Rabin, and adapted from [256].



Figure 5.4: Interpolation between three different shapes (a five-pointed star, a four-pointed one, and the union of two small disks). Every image represents a barycenter between the three shapes, computed in $\mathbb{W}_2$. Pictures kindly provided by G. Peyré.

Chapter 6

Benamou-Brenier and other continuous numerical methods

In this chapter we present some numerical methods to solve optimal transport problems. The search for efficient numerical strategies is a very lively domain in current days, after a period where optimal transport was mainly investigated from the theoretical point of view. It would be impossible to list all the recent developments in the field and, because of the taste for Calculus of Variations and PDEs which haunts all the book, the choice has been done to prefer those methods which can be classified as “continuous”. This means that they are especially suitable to tackle the problem of the optimal transport between two given densities. They are opposite to some more discrete methods, that will be briefly presented in the discussion section 6.4, and are concerned with the case of two finitely atomic measures or one atomic vs one density. The most famous method is for sure the one due to J.-D. Benamou and Y. Brenier, which transforms the problem into a tractable convex variational problem in dimension $d + 1$. This method gives the title to the chapter, and justifies the position of this chapter after Chapter 5. Indeed, we will describe it strongly using the theory about Wasserstein geodesics (rather than finding the map, this method finds the geodesic curve μ_t).

Two other classical continuous methods are presented: the Angenent-Haker-Tannenbaum method based on the fact that the optimal maps should be a gradient, and that removing non-gradient parts decreases the energy, and the Loeper-Rapetti method based on the resolution of the Monge-Ampère equation. Both require smooth and non-vanishing densities, and special domains to handle boundary data (a rectangle or, better, a torus). The problem of the resolution of the Monge Ampère equation on more general domains is much more delicate and has been addressed more recently (see the end of Section 6.3).

In this chapter we will neither be exhaustive, nor present the most recent methods or compare the efficiency and advantages of all of them. It would be impossible, because of the huge progresses which are occurring on this topic. The goal is mainly to provide the reader an overview of the most natural ideas which could be translated into numerical tools.

6.1 The Benamou-Brenier formula and its numerical applications

The results of Sections 5.3 and 5.4 allow to rewrite the optimization problem corresponding to the cost $|x - y|^p$ in a smart way, so as to transform it into a convex optimization problem. Indeed

- looking for an optimal transport for the cost $c(x, y) = |x - y|^p$ is equivalent to looking for constant speed geodesic in $\mathbb{W}_p$ because from optimal plans we can reconstruct geodesics and from geodesics (via their velocity field) it is possible to reconstruct the optimal transport;
- constant speed geodesics may be found by minimizing $\int_0^1 |\mu'(t)|^p dt$;
- in the case of the Wasserstein spaces, we have $|\mu'(t)|^p = \int_{\Omega} |\mathbf{v}_t|^p d\mu_t$, where $\mathbf{v}$ is a velocity field solving the continuity equation together with μ (this field is not unique, but the metric derivative $|\mu'(t)|$ equals the minimal value of the L^p norm of all possible fields).

As a consequence of these last considerations, for $p > 1$, solving the kinetic energy minimization problem

$$\min \left\{ \int_0^1 \int_{\Omega} |\mathbf{v}_t|^p d\rho_t dt : \partial_t \rho_t + \nabla \cdot (\mathbf{v}_t \rho_t) = 0, \rho_0 = \mu, \rho_1 = \nu \right\}$$

selects constant speed geodesics connecting μ to ν , and hence allows to find the optimal transport between these two measures¹.

On the other hand, this minimization problem in the variables $(\rho_t, \mathbf{v}_t)$ has non-linear constraints (due to the product $\mathbf{v}_t \rho_t$) and the functional is non-convex (since $(t, x) \mapsto t|x|^p$ is not convex). Yet, we saw with the tools of Section 5.3.1 that it is possible to transform it into a convex problem.

For this, it is sufficient to switch variables, from $(\rho_t, \mathbf{v}_t)$ into (ρ_t, E_t) where $E_t = \mathbf{v}_t \rho_t$ and use the functional $\mathcal{B}_p$ in space-time. Remember $\mathcal{B}_p(\rho, E) := \int f_p(\rho, E)$ with $f_p : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ defined in Lemma 5.17. We recall its definition for the reader's convenience:

$$f_p(t, x) := \sup_{(a, b) \in K_q} (at + b \cdot x) = \begin{cases} \frac{1}{p} \frac{|x|^p}{t^{p-1}} & \text{if } t > 0, \\ 0 & \text{if } t = 0, x = 0 \\ +\infty & \text{if } t = 0, x \neq 0, \text{ or } t < 0, \end{cases}$$

where $K_q := \{(a, b) \in \mathbb{R} \times \mathbb{R}^d : a + \frac{1}{q}|b|^q \leq 0\}$. The problem becomes:

Problem 6.1. Solve

$$(\mathbf{B}_p \mathbf{P}) \quad \min \left\{ \mathcal{B}_p(\rho, E) : \partial_t \rho_t + \nabla \cdot E_t = 0, \rho_0 = \mu, \rho_1 = \nu \right\}.$$

¹We choose to denote by ρ the interpolating measures, in order to stress the “continuous” flavor of this method (typically, ρ is a density, and μ a generic measure)